From the original source to the filled geometric saddle

Original positive endpoint source

$$Z_{D,N} \quad = \quad [\prod_{f \text{ ordinary}} L_f^{N-1}] \prod_{e=1}^{R} P_e\big|_{L_u = L_v = 0}$$

Original faces, marked targets and level N retained.

Full polynomial support

Every local degree-$N-1$ monomial

Exact integer incidence flows

One compact flow polytope $\mathcal{Q}_D$

Uniform coefficient entropy

$h(q_e)$ on the entire closed simplex

Cross and one-sided branches agree at their terminal seams

Proved positive-source limit at every fixed plat width

$$\lim_{N\to\infty} \frac{\log Z_{D,N}}{N} \quad = \quad W_D \quad = \quad \max_{q\in\mathcal{Q}_D} \sum_e h(q_e)$$

Even endpoint; odd levels use the actual spectral regularization.

Proved under the original labeled geometric certificate

$$W_D \quad = \quad \frac{\mathrm{Vol}(Y_D)}{2\pi}$$

Exact identity contraction, hybrid flow cuts, and the relative fundamental chain of the original augmented-link group.

Proved: full complex spectral source near completeness

Original face torus, nonzero relative asymptotic, all repeated-node spectral data retained.

The unit-charge filling critical value is $\mathrm{Vol}(S^3 \setminus K_D(p))/(2\pi)$.

Proved: the localized original spectral integral

Nonzero unit-charge contribution and finite local charge dominance for independently long, fixed signed twists.

$$\lim_{N\to\infty} \frac{2\pi}{N} \log |U_{N,n;b}| \quad = \quad \mathrm{Vol}(S^3 \setminus K_D(p))$$

Open: control the exact global remainder

Selected charges outside the cube; every other charge on its full domain. Their total must not cancel or dominate the proved local contribution.